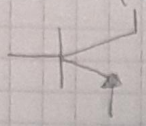
Sıra çelinkol, Eres sağlıcan

physics of semiconductor devices Sze s. m 2ed edition

Ders içeriği: Katı Yapılar, Elektron için hareketler, yalıtkan iletken yarı iletken fiziği

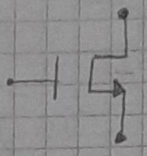
1- Diyot → (PN Junction)

2) BJT (Bipolar Junction transistor)



3) FET (Alan Field etkili effect transistor)

JFET (Junction) MOSFET →



AC → DC

Kaynaklar:

Ders kitabı: ★ Microelectronic Circuits → 2014
↳ Sedra Smith) The oxford series in electrical and computer engineering

Elektronik → mehmet sait Türköz → Birsen yayınevi, 2004

Değerlendirme
↳ 10 ödev, %5, 1 Proje → LTSPICE

Vize → %30
Final → %50

Detail

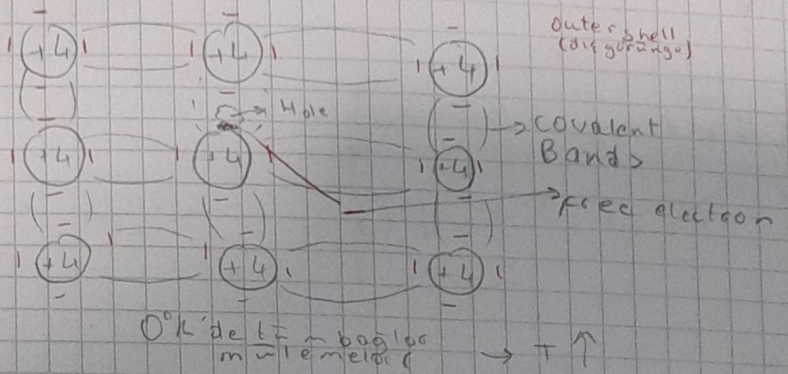
Semiconductors

- Intrinsic Semiconductors (Has yarıiletken)
- Doped Semiconductor (katkılı ")

→ Elemental Semi → Silicon
→ Germanyum

→ GrAs → Gallium Arsenide
→ comp and semi

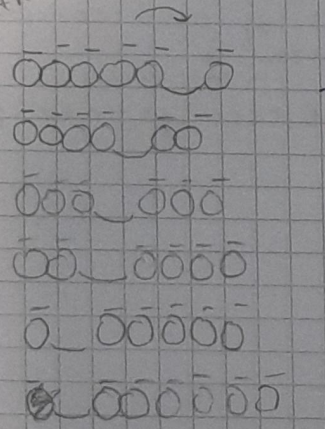
Intrinsic Hass yarıiletken
Silicon



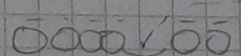
0°K'de tüm bağlar mükemmeldir → T ↑

Hole (Elektron boşluğu)

Boşluk hareketi ?

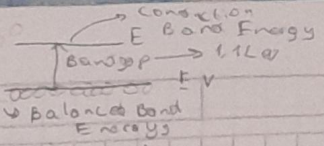


Recombination (Rekombinasyon)



free electron anhilate : serbest elektronun boşlukla yer değiştirmesi

Termal dengede (Hass yarıiletkende)
Thermal generation = Recombination
sıcaklıkla üretilen yeni boş elektron sayısı



Free carriers
(Serbest Taşıyıcı)

electrons (Elektron)

holes (Boşluk)

(Has yarı iletken)

$n = p = n_i$

elektron sayısı — holes (Boşluk) sayısı → Band gap Energy (Bant aralığı ")

$n_i = \beta T^{3/2} e^{-Eg/2kT}$

Paramatre (silikon) $= 7,3 \cdot 10^{15} cm^{-3} k^{-3/2}$

$T = 300 k$

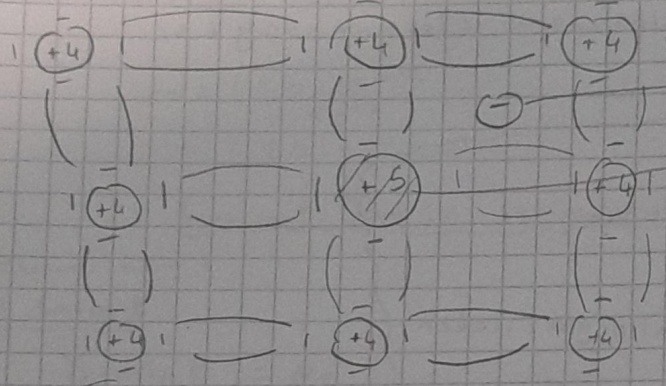
$\Rightarrow n_i = 1,5 \times 10^{10}$ carrier/cm^3

silicon → $5 \cdot 10^{22}$ atom/cm^3

Doped (katkılı) semiconductor

$n \cdot p = n_i^2$

↳ Has yarıiletken ve katkılı yarıiletken için geçerli.



Doped semiconductor

n-tipi yarıiletken | p-tipi yarıiletken

Çoğunluk taşıyıcıları elektronlar

Çoğunluk taşıyıcıları boşluk

$P_n \cdot n_n = n_i^2$

yarıiletkenin tipi

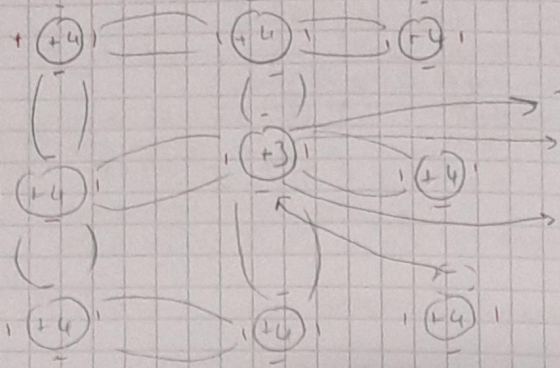
intrinsic

n tipi

P_n : n-tipi yarıiletkende boşluk sayısı

n_n : " " elektron sayısı

P tipi



N_A : p tipi yarı iletken

N_D Donor Atoms (+5 valance electron)

n tipi yarı iletken

$n_p \cdot p_p = n_i^2$

p_p : p-tipi yarıiletkendeki boşluk sayısı

n_p = p tipi yarıiletkendeki elektron sayısı

$p_n \cdot n_n = n_i^2$

$N_D \rightarrow$ n-tipi yarıiletken

$n_n \approx N_D$

$p_n = \frac{n_i^2}{N_d}$

n tipi yarıiletken

p-tipi yarıiletken

$P_p \approx N_A$ → Acceptor Atom yoğunluğu

$P_p n_p = n_i^2 \Rightarrow n_p \approx \frac{n_i^2}{N_A}$

Elektron sayısı

	n-tipi	p-tipi
Çoğunluk taşıyıcılar (Majority carriers)	electrons	holes
Azınlık taşıyıcılar (minority carriers)	holes	electrones

Example 3.2
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n-type silicon, katkı yoğunluğu (Doopant concentration) $N_D = 10^{17}/cm^3$ electron ve boşluk yoğunluğu T=300k bulunuz

$$n_n = N_D \approx 10^{17}/cm^3$$

$$p_n = \frac{n_i^2}{N_D} \Rightarrow n_i = 1,5 \cdot 10^{10}/cm^3$$

$$p_n = \frac{(1,5\ 10^{10})^2}{10^{17}}$$

$$= 2,25 \cdot 10^3/cm^3$$

Current Flow in semiconductors

(sürükleme akımı)

Drift Current

$v_{p\text{-drift}}$ (hız) $= \mu_p \cdot E$ (mobilite hole) → Elektrikalan

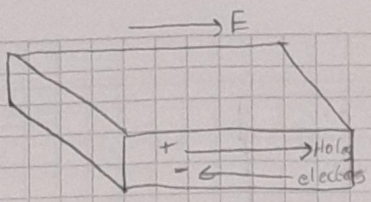
$\mu_p = 480 cm^2/V.s$

$v = \mu \cdot E$ → Elektr. alan (hız)

$$\frac{cm}{s} = \mu \cdot \frac{V}{cm} \Rightarrow \frac{cm^2}{V.s}$$

$v_{n\ drift} = -\mu_n \cdot E$

Elektron mobilitesi → $1350 cm^2/V.s$



in one second

$I_p = A \cdot q \cdot p \cdot (v_{p\,drift})$

$A \cdot q \cdot p \cdot v \rightarrow C$

delikler için $I_p = A \cdot q \cdot p \cdot \mu_p \cdot E$

$\rightarrow J \rightarrow$ current density

$J_p = \frac{I_p}{A} = q \cdot p \cdot \mu_p \cdot E$

elektronlar için $I_n = -A \cdot q \cdot n \cdot v_{n\,drift} \rightarrow -\mu \cdot E$

$J_n = \frac{I_n}{A} = q \cdot n \cdot \mu_n \cdot E$

$J = J_p + J_n = q(p\mu_p + n\mu_n)E$

$J = \sigma \cdot E$

σ conductivity (iletkenlik)

ρ resistivity $= \frac{1}{\sigma} = \frac{1}{q(p\mu_p + n\mu_n)}$

$\rho = \frac{E}{J}$ $\qquad \rho = R \frac{A}{\ell}$ → kesit alanı → uzunluk

Example 3.3

Resistivity = ?

a) intrinsic Silicon

b) p-type $\approx \rightarrow N_A = 10^{16}/cm^3$

$n_i = 1,5 \cdot 10^{10}/cm^3$

has yarıiletken $\longrightarrow \mu_n = 1350\ cm^2/V.s$

$\mu_p = 470 cm^2/Vs$

$\rightarrow$ katkılı yarıiletken $\rightarrow \mu_n = 1110 cm^2/Vs$

$\rightarrow \mu_p = 400 cm^2/Vs$

$p = n = n_i = 1,5 \cdot 10^{10}/cm^3$

a) $\rho = \frac{1}{q(p\mu_p + n\mu_n)} = \frac{1}{1,6 \cdot 10^{-19}(1,5 \cdot 10^{10} \cdot 480 + 1,5 \cdot 10^{10} \cdot 1350)}$

$= 2,28 \times 10^5\ \Omega\ cm \rightarrow$ Has yarıiletken

b) p tipi yarıiletken

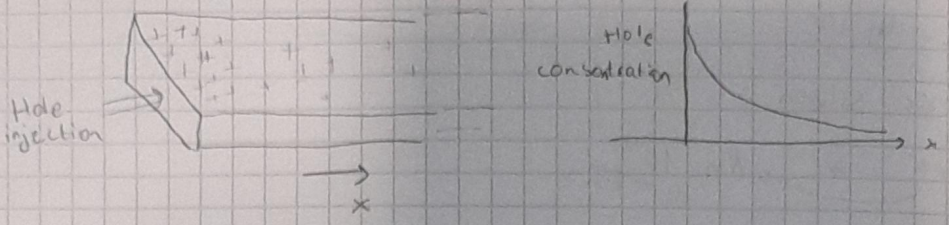
$p_p \simeq N_A = 10^{16}$

$n_p \simeq \frac{n_i^2}{N_A} = \frac{(1,5 \cdot 10^{10})^2}{10^{16}} = 2,25 \cdot 10^4/cm^3$

$\rho = \frac{1}{q(p\mu_p + n\mu_n)} = \frac{1}{1,6 \times 10^{-19}(10^{16} \times 400 + 2,25 \times 10^4 \times 1110)}$

$\simeq \frac{1}{1,6 \times 10^{-19} \times 10^{16} \cdot 400} = 1,56\ \Omega\ cm$

Diffusion Current

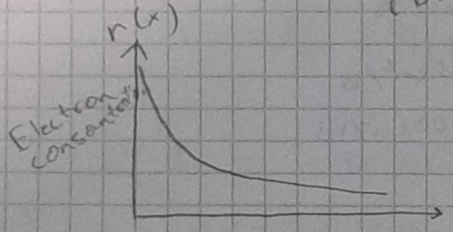


$$J_p = -q D_p \frac{dp(x)}{dx}$$

difüzyon sabiti (Boltzmann)

$$J_n = q D_n \frac{dn(x)}{dx}$$

(D_n: elektron difüzyon sabiti)



Diffusion constans

↳ $D_p = 12\ cm^2/s$

$D_n = 35\ cm^2/s$

Example 3.4

$p(x) = p_0 e^{-x/L_p}$

Find the hole donsity at $x = 0$

$p_0 = 10^{16}/cm^3$, $L_p = 1\mu m$, $D_p = 12\ cm^2/s$

$I_p = ?$ (Bar cross sectional Area $10\ \mu m^2$)

$$J_p = -q D_p \frac{dp(x)}{dx}$$

$$= -q D_p \frac{d}{dx}\left[p_0 e^{-x/L_p}\right]$$

$$= q \frac{D_p}{L_p} p_0 e^{-x/L_p}$$

$$J_p(0) = q \frac{D_p}{L_p} p_0$$

$$= 1,6 \cdot 10^{-19} \cdot \frac{12}{1 \cdot 10^{-4}} \cdot 10^{16} = 192\ A/cm^2$$

$I_p = J_p \times A \Rightarrow 192 \cdot 100 \frac{1 \cdot 10^{-4}}{10^{-2}} = 192\ \mu A$

Relationship between D and M

Einstein relationship

$$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = V_T$$

V_T → Termal Gerilim / Thermal Voltage → $V_T = \frac{kT}{q}$

$T = 300K \Rightarrow V_T \approx 25.9mV$

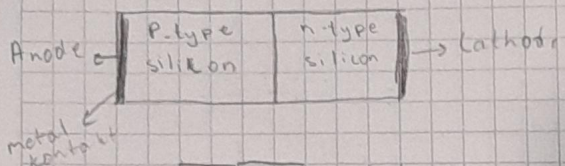
$\approx 26mV, 25mV$

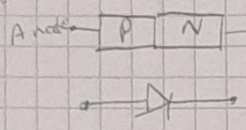
2. ders

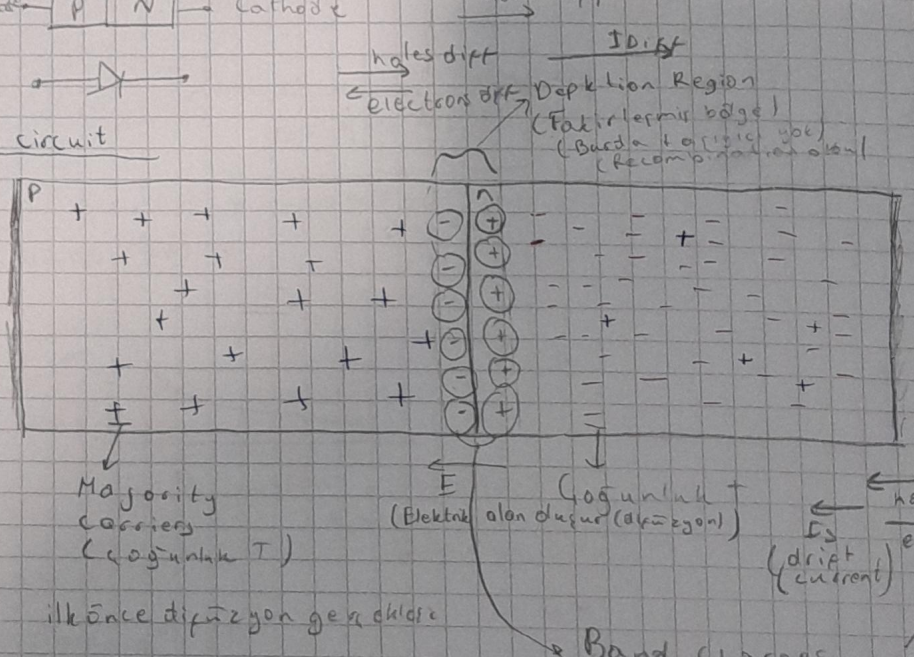
PN Junction'ım

$n_i = 1.5 \cdot 10^{10}$ silikonda

$n \cdot p = n^2$







ilkönce difüzyon gerçekleşir

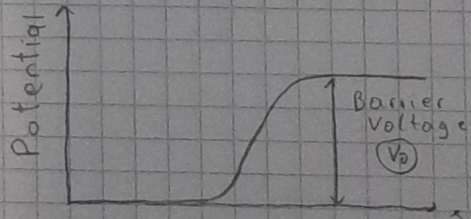
Artıyükler elektrik alanla aynı yönde eksiler ise ters yönde

Dengede ⇒ $I_D = I_S$

I_D → difüzyon akımı

I_S → drift current

Elektrik alan nedeniyle



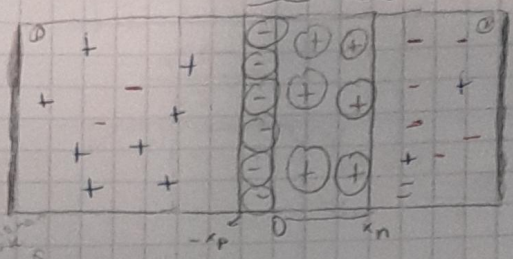
$$V_0 = V_T \ln\left(\frac{N_A N_D}{n_i}\right)$$

→ Dengede olan (+) type

N_A → Acceptor atom (+3) P type

Farklı katkı yoğunluklarında pn jonksiyonu

$N_A > N_D$



$$\frac{x_n}{x_p} = \frac{N_A}{N_D}$$

$|Q_+| = q A x_n N_D$

$|Q_-| = q A x_p N_A$

$\rightarrow q A x_n N_D = q A x_p N_A$

$$\frac{x_n}{x_p} = \frac{N_A}{N_D}$$

x_n : n tipi bölgedeki yakiclaşmış bölgenin uzunluğu

x_p : p tipi " " " "

W (the width of depletion region) $= x_n + x_p = \sqrt{\frac{2\varepsilon_s}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right) V_0}$

$\varepsilon_s = \varepsilon_r \cdot \varepsilon_0$

$= (11.7)(8{,}85 \cdot 10^{-14}\ F/cm)$

$\frac{x_n}{x_p} = \frac{N_A}{N_D} \Rightarrow x_p = x_n \frac{N_A}{N_D}$

$x_n - x_p = W$

$x_n + x_n \frac{N_D}{N_A} = W$

$x_n \left(1 + \frac{N_D}{N_A}\right) = W$

$x_n = \frac{N_A}{N_A + N_D} \cdot W$

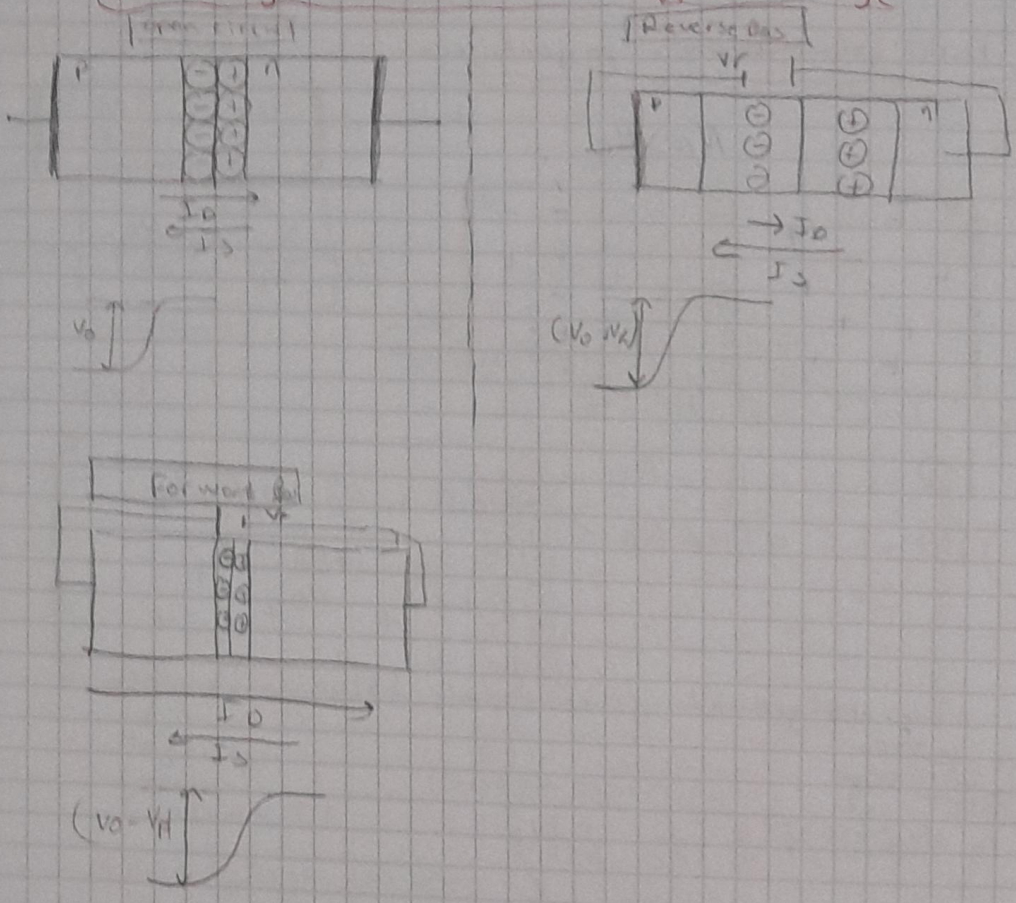
$x_p = W \frac{N_D}{N_A + N_D}$

$|Q_+| = |Q_-| = Q_J$

$Q_J = A \cdot q \left(\frac{N_A N_D}{N_A + N_D}\right) W$

$$Q_J = A \sqrt{2\varepsilon_s q \left(\frac{N_A N_D}{N_A + N_D}\right) V_0}$$

The PN junction with an Applied Voltage



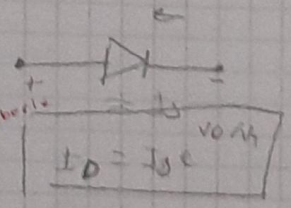
Reverse

$$W = x_n + x_p = \sqrt{\frac{2\varepsilon_s}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)(V_0 + V_R)}$$

$$Q_J = A\sqrt{2\varepsilon_s q\left(\frac{N_A N_D}{N_A + N_D}\right)(V_0 + V_R)}$$

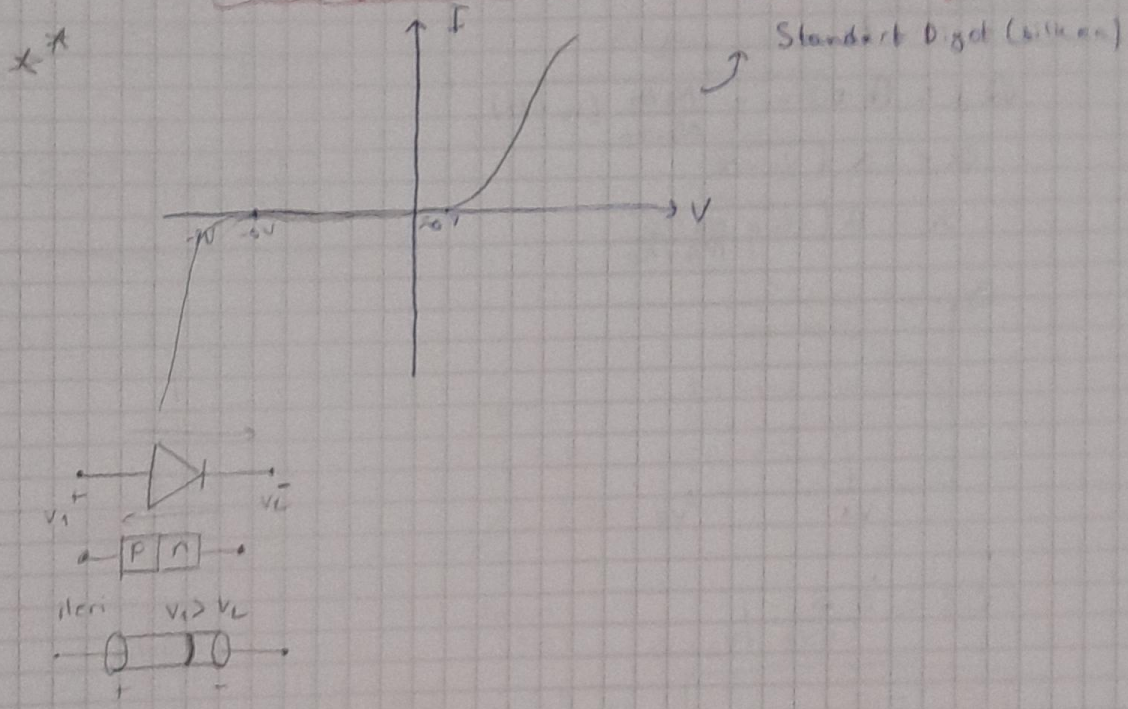
Reverse $\rightarrow I = I_S$

Forward $\rightarrow \boxed{I = I_S\left(e^{V/V_T} - 1\right)}$



I_S sıcaklıktan çok etkilenir

The PN Junction with an applied Voltage



Reverse Breakdown

→ Zenner effect → elektrik alanıyla bağların kopması
→ Avalanche effect

Capacitive effects

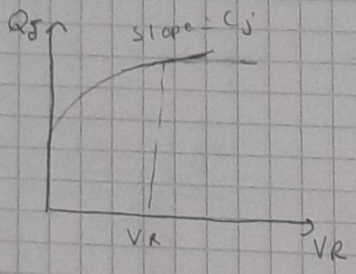
→ Depletion of junction capacitance

↳ Reverse region

$$Q_J = A\sqrt{2\varepsilon_s q \frac{N_A N_D}{N_A + N_D}(V_0 + V_R)}$$

$$Q_J = \alpha\sqrt{V_0 + V_R}$$

$$\alpha = A\sqrt{2\varepsilon_s q \frac{N_A N_D}{N_A + N_D}}$$



$$C_j = \frac{d Q_J}{d V_R}\Big|_{V_R = V_0}$$

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$$\boxed{C_j = \frac{\alpha}{2\sqrt{V_0 + V_R}}} \longrightarrow C_j = \frac{C_{j0}}{\sqrt{1 + \frac{V_R}{V_0}}}$$

$C_j0 \rightarrow (V_R = 0)$

Difuzyon kapasitesi
(Diffusion capacitance)

→ Forward Bias

↳ minority carriers in effect

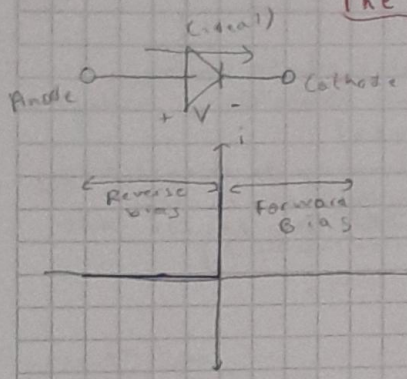
$$C_d = \left(\frac{\tau_T}{V_T}\right) I$$

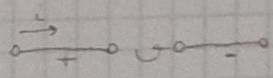
τ_T → mean transit time

C_d ← incremental diffusion capacitance

>ojt a 169-170
Sezai ok

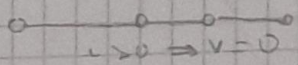
The ideal Diode





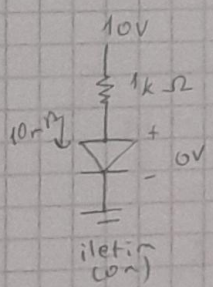
$V < 0 \Rightarrow i = 0$

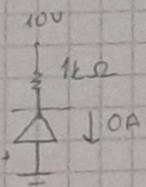
Reverse (off)
(Ters yön)
(kesim)

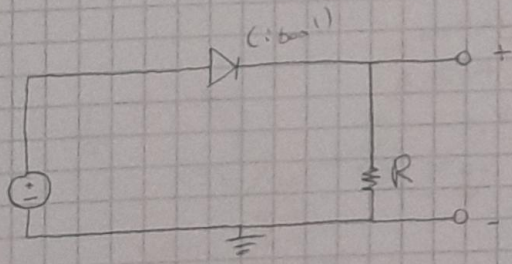


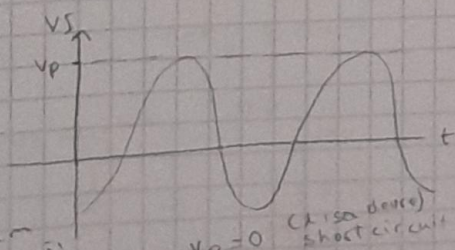
$i > 0 \Rightarrow V = 0$

Forward (on)
(ileri yön)
(iletim)

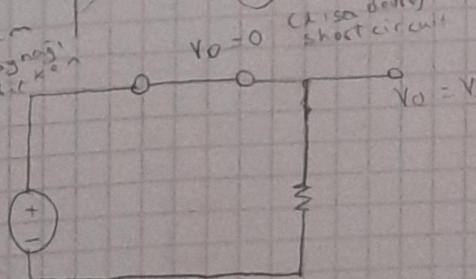




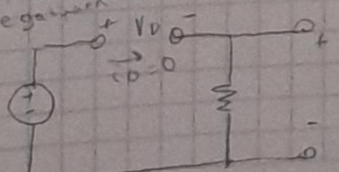


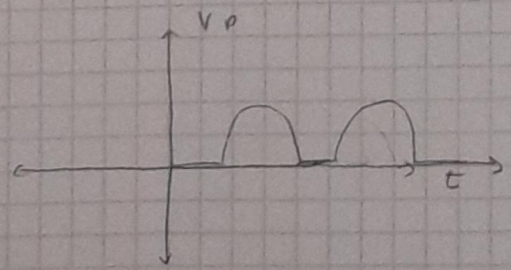


1. durum
Giriş kaynağı pozitif çevrim

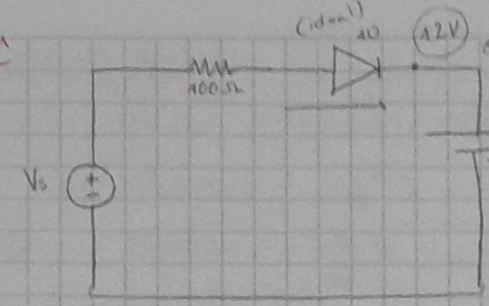


Giriş kaynağı negatifken





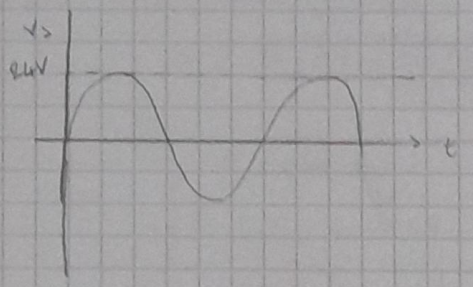
Example 4.1



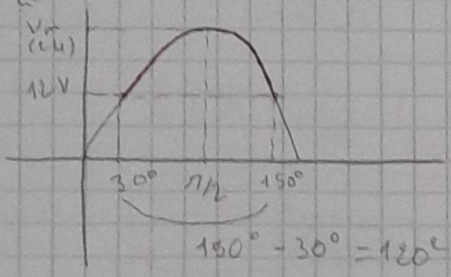
a)- Peak value of diode current

b)- Maximum reverse-bias voltage on the diode?

c)- Fraction of each cycle during which the diode conducts (periodun kaçta kaçında diot iletimdedir?)



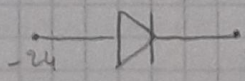
c) Çözüm



$150° - 30° = 120° \rightarrow 1/3$ cycle

a) $I_D = \frac{24-12}{100} = 0,12\,A$

b)

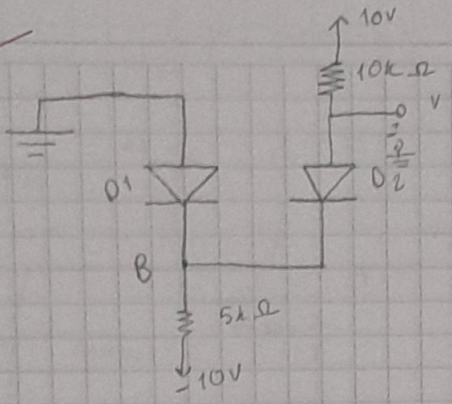


$-24 - 12 = -36V$

Katot orasına diyen negatif gerilim

Logic Gates

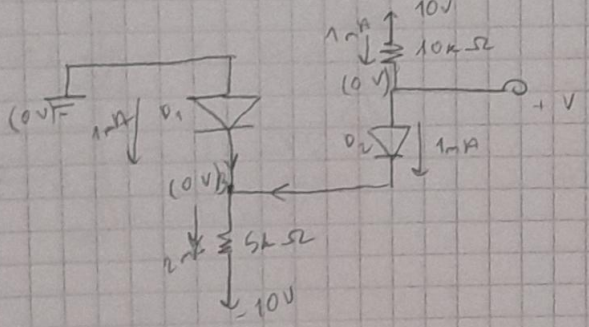
Örnek 2



a)

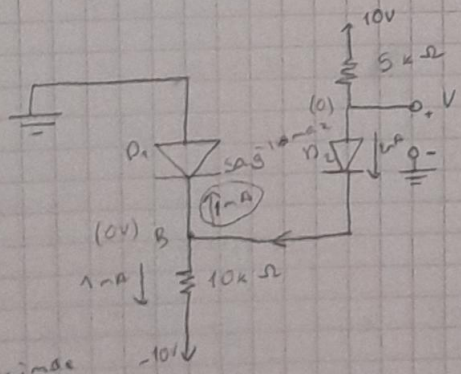
Gözlemlerimiz varsayımda bulunduk

D_1 is on and D_2 is on ✓ ✓



iletimdeki akım akar ve pota kısa devre gibi davranır.

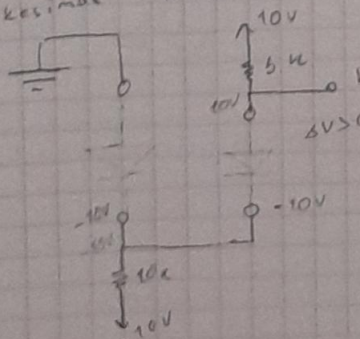
b)



yorum

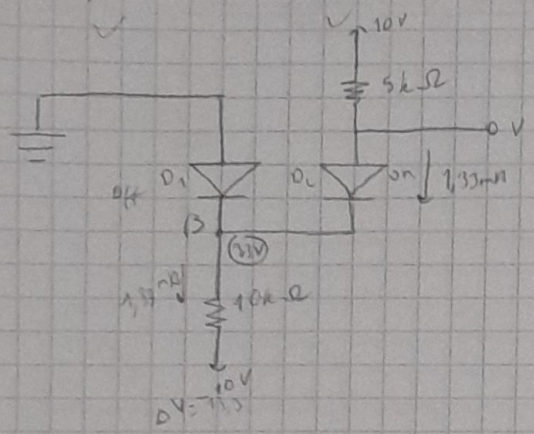
D_1 is on and D_2 is on varsayımı yanlış

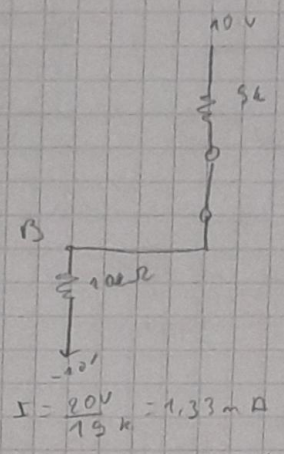
D_1 ve D_2 kesimde



$\Delta V > 0$ hipotez yanlış

D_1 is off and D_2 is on

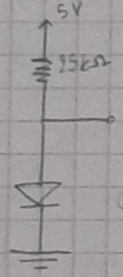




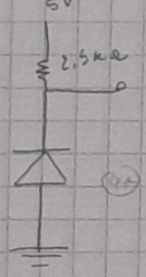
$I = \frac{20V}{15k} = 1,33mA$

Exercise 4.4 p. 183

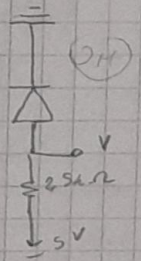
(a)



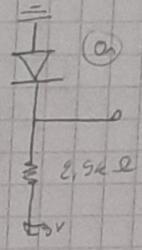
(b)

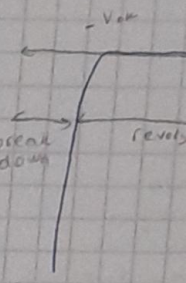


(c)

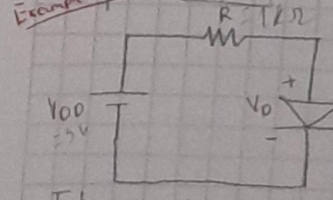


(d)

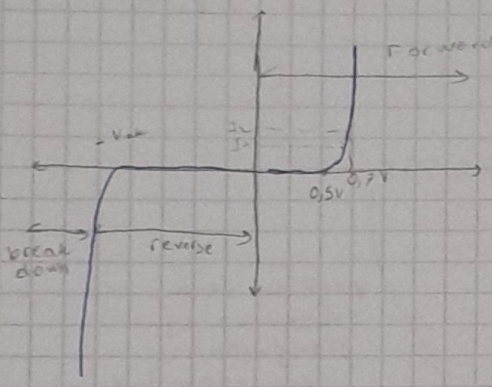




Example 4.4

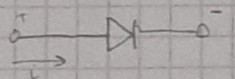


4.2 Terminal Characteristics of Junction Diodes



The Forward - Bias Region

$$i = I_S \left(e^{v/V_T} - 1 \right)$$



$$i \approx I_S e^{v/V_T}$$

$$v = V_T \ln \frac{i}{I_S}$$

2 değerlerden $I_1 = I_S e^{v_1/V_T}$

$I_2 = I_S e^{v_2/V_T}$

$$\frac{I_2}{I_1} = e^{(v_2 - v_1)/V_T}$$

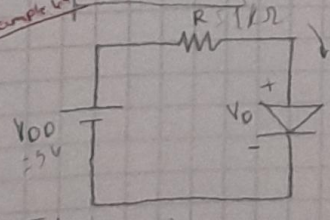
$$v_2 - v_1 = V_T \ln \frac{I_2}{I_1}$$

$$v_2 - v_1 = (e\, V_T) \log \frac{I_2}{I_1} = \underset{60mV}{(2,3 V_T)} \log \frac{I_2}{I_1}$$

Wafer, IC process, IC package

$i \approx -I_S$ → Reverse Region

Example 4.4



→

ideal diyot modeli
$V_D = 0\,V$
$I_D = \frac{5 - 0}{1} = 5\,mA$

The diode current is 1mA @ 0,7V

constant Voltage drop model
(sabit gerilim düşümlü model)
$V_D = 0,7V$

$$I_D = \frac{5 - 0,7}{1} = 4,3\,mA$$

Exponantial model

$$I_D = \frac{V_{DD} - V_D}{R} \quad (1) \qquad V_2 - V_1 = 2{,}3\, V_T \log \frac{I_2}{I_1} \quad (2)$$

$$I_D = \frac{5 - 0{,}7}{1} = 4{,}3 \text{ mA}$$

$$V_2 = V_1 + 0{,}06 \log \frac{I_2}{I_1} \Rightarrow V_2 = 0{,}738$$

$V_1 = 0{,}7V$, $I_1 = 1mA$ ⇑ $I_2 = 4{,}3mA$

Iteration 2nd

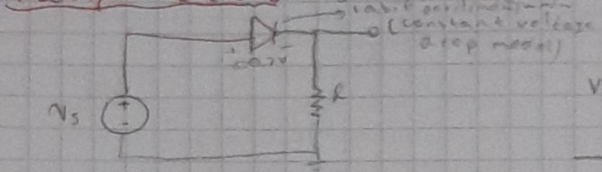
$$I_D = \frac{5 - 0{,}738}{1} = \boxed{4{,}262 \text{ mA}}$$

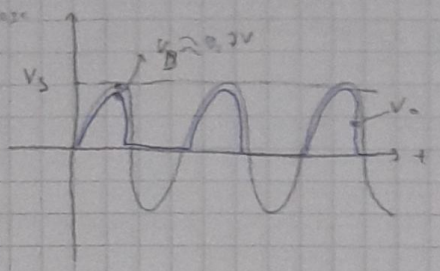
$$V_2 = 0{,}738 + 0{,}06 \left[\frac{4{,}262}{4{,}3} \right]$$

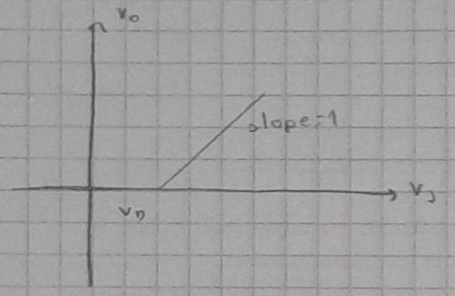
$$= \boxed{0{,}738 \text{ V}}$$

virgülden sonra 3 basamak aynı olduğundan iterasyonu durdurduk

Rectifier circuits (Doğrultucu devre)

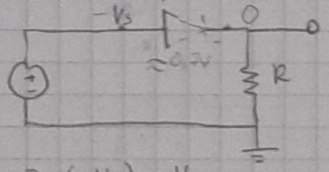




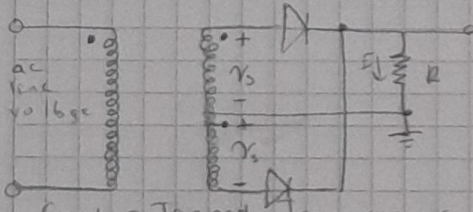


PIV. Peak inverse Diod voltage (Diyotlara gerilimin tepe değeri)

PIV: V_s ⇒ nasıl bulunur, diot kesimdeiken



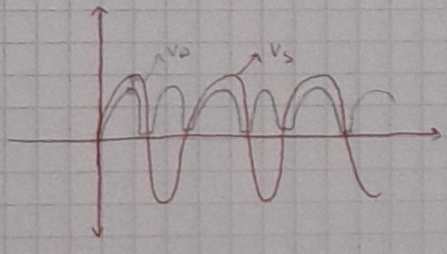
$PIV = 0-(-V_s) = V_s$



D_1 D_2

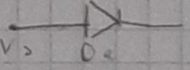
In positive cycle on off

In negative cycle off on



PIV:

In positive cycle D_2 is off

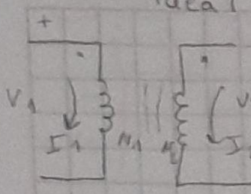


$PIV = V_o - (-V_s) = V_o + V_s$

$V_o = V_s - V_D$

$PIV = (V_s - V_D) + V_s = 2V_s - V_D$

ideal

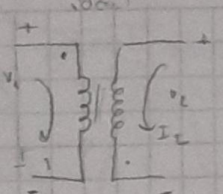


N_1 ve N_2 sarım sayıları

$$\frac{V_1}{N_1} = -\frac{V_2}{N_2}$$

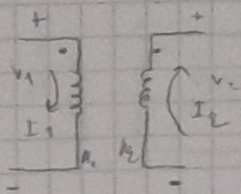
$$N_1 I_1 = -N_2 I_2$$

ideal



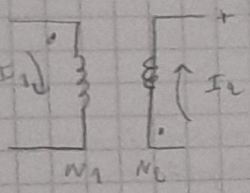
$$\frac{V_1}{N_1} = -\frac{V_2}{N_2}$$

$$N_1 I_1 = -N_2 I_2$$



$$\frac{V_1}{N_1} = \frac{V_2}{N_2}$$

$$N_1 I_1 = N_2 I_2$$



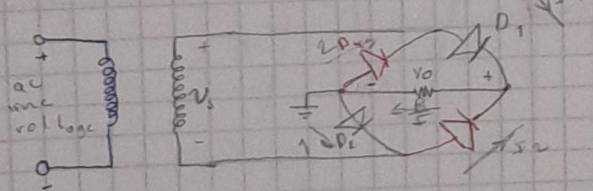
$$\frac{V_1}{N_1} = -\frac{V_2}{N_2}$$

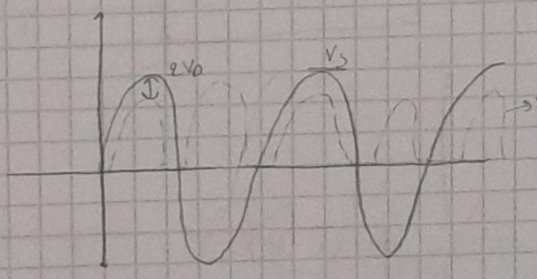
$$N_1 I_1 = -N_2 I_2$$

noktalar aynı yerde ise 1. denklem artı

akımlar aynı noktadan sıkı yerse 2. -

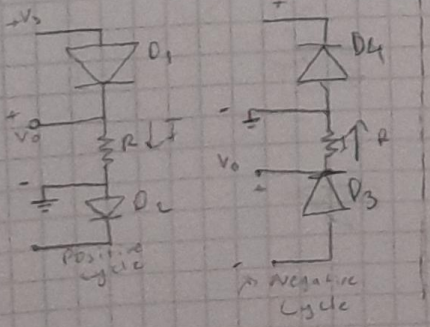
The bridge rectifier

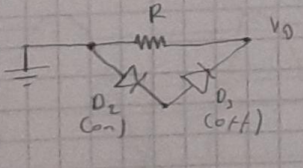




Full wave

	D_1	D_2	D_3	D_4
Positive Cycle	on	on	off	off
Negative Cycle	off	off	on	on



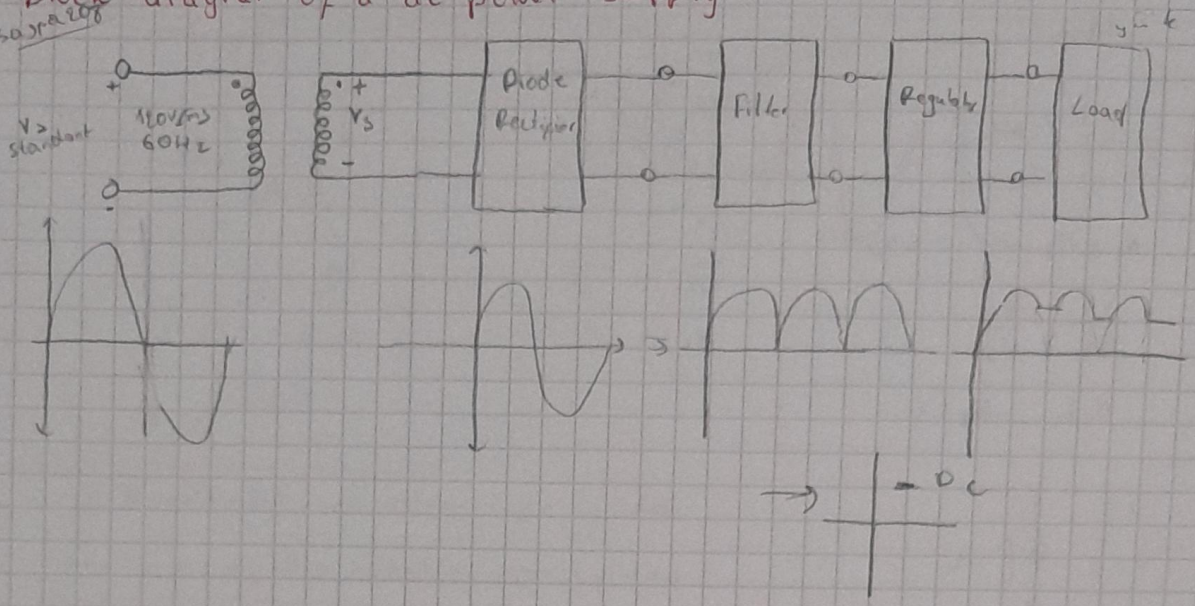


$$V_{D_3} = V_o - (-V_{D_2})$$
$$= V_o + V_{D_2}$$

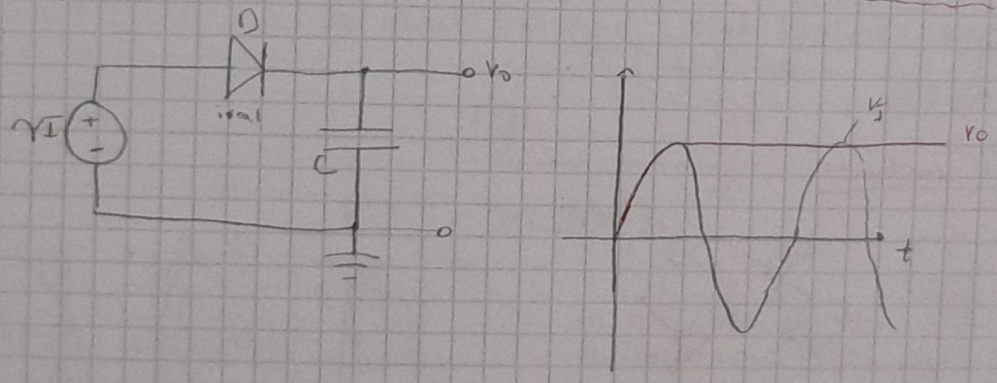
$$V_o \text{ peak} = V_s - 2V_D \qquad V_{D_2} = V_D$$

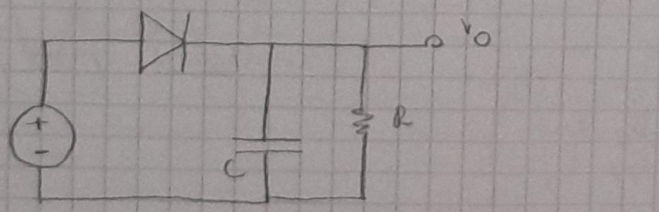
$$\Rightarrow V_{D_3}(\text{reverse}) = (V_s - 2V_D) + V_D$$
$$= V_s - V_D$$

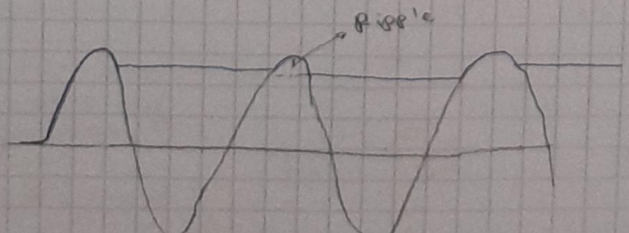
Block diagram of a dc power supply

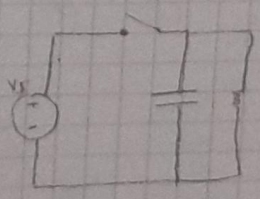


The rectifier with a Filter Capacitor - The peak rectifier



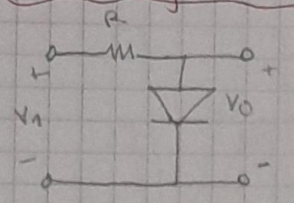


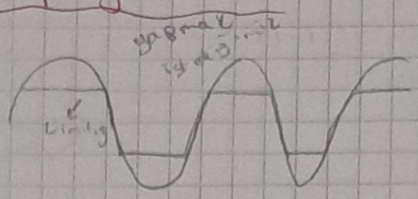


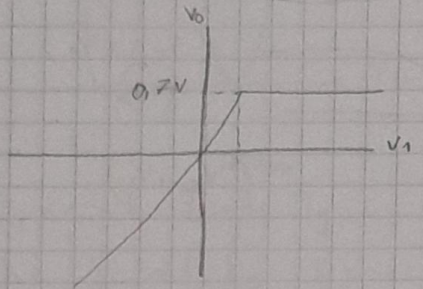


$V_r = \frac{V_p}{fCR}$ → giving us the ripple

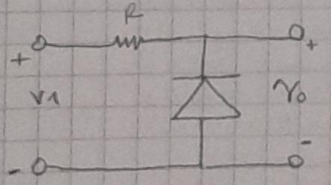
Limiting and Clamping Circuit

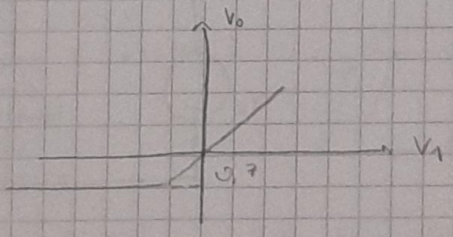


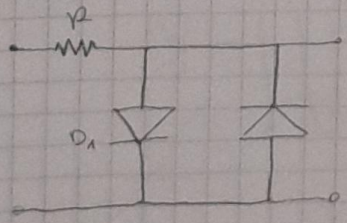


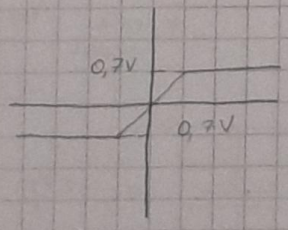


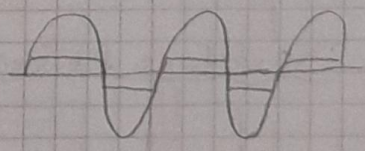
Limiting occures when the diodes are conducting

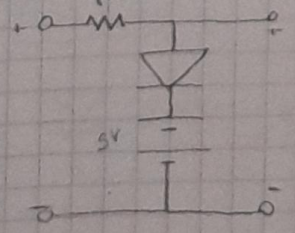


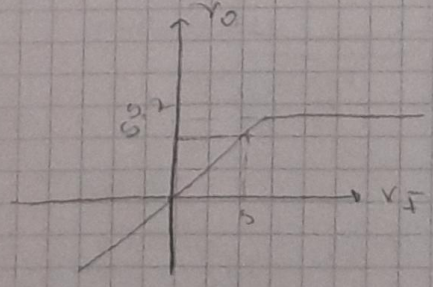




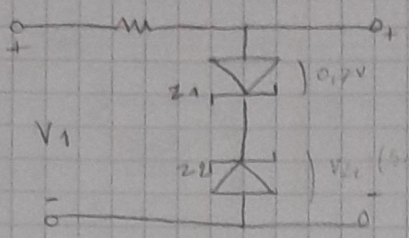


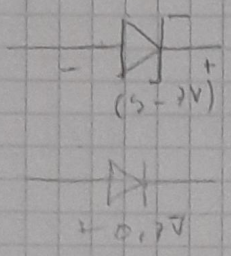


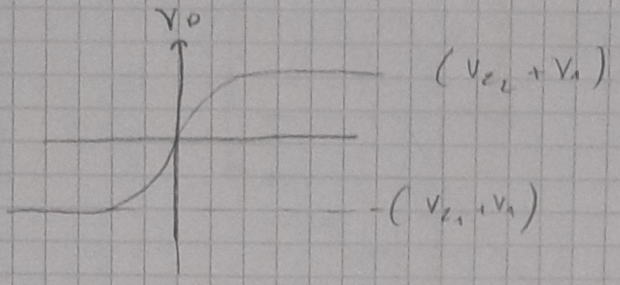




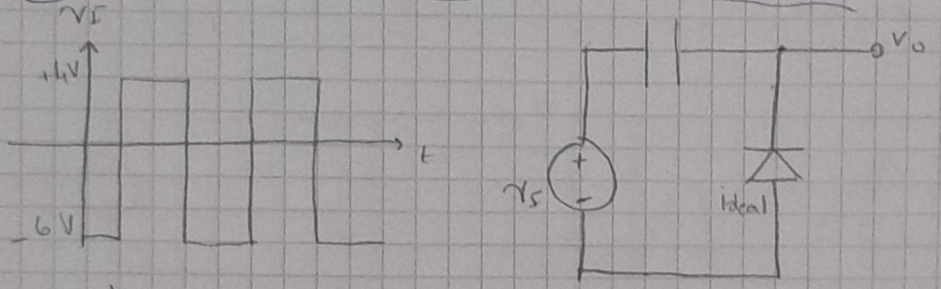
zener diyot diyotun tanitisi

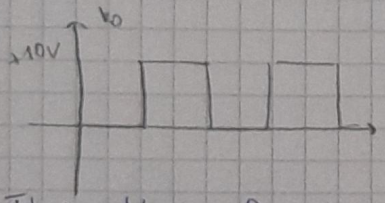


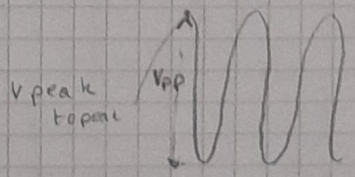




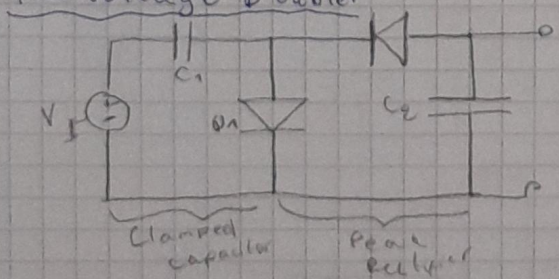
The clamped Capacitor or DC restroner

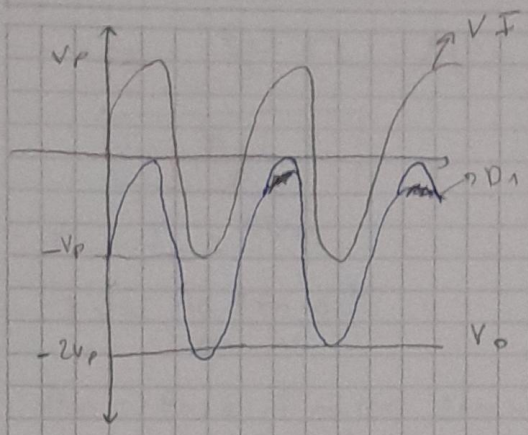






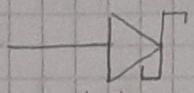
The voltage Doubler



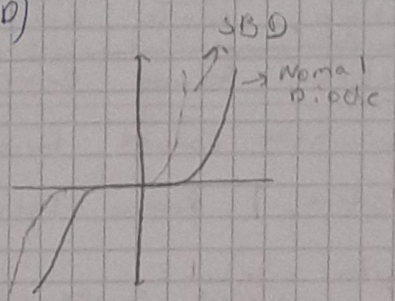


Special Diode Types

- The Schottky Barier Diode (SBD)

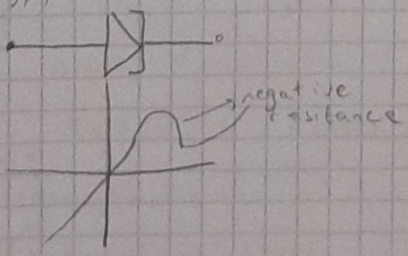


- metal semiconductor
- voltage drop 0.3V - 0.5V
- faster switching (speed)
  - power supply
  - (clamped mode)
  - single-hall voltage clamping

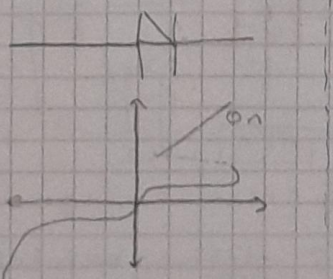


Tunnel Diode (Esaki Diode)

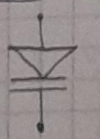
(1957)



Shockley Diode

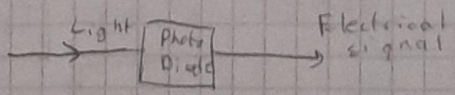


Varactors

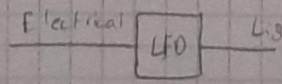


→ Voltage variable capacitors

Photo Diode



LED

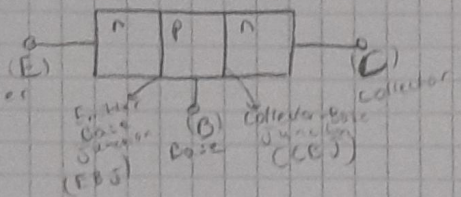


Optoizolator

Bipolar Junction Transistor (BJT)

akım taşıyıcıları iki tür: elektronlar ve boşluk

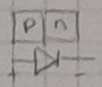
Transistör

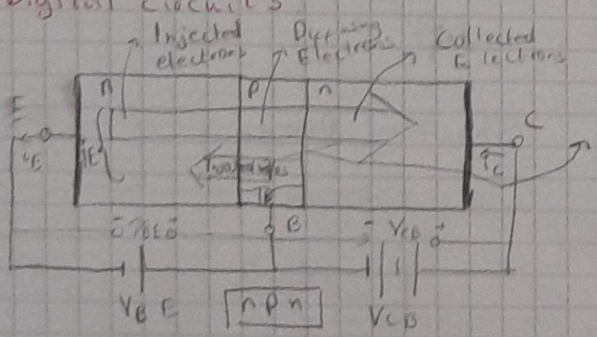


BJT Modes of Operation

Mode	EBJ	CBJ
Cut off	Reverse	Reverse
Active	Forward	Reverse → amplifier
Saturation	Forward	Forward
(Reverse active)	Reverse	Forward

Digital circuits





Elektrik alanın etkisiyle $I_E > I_C$

$V_{CB} > V_{BE}$

p bölgesinde recombinasyonu önlemek için çok küçük yaparım)

akım yönü elektron hareketiyle ters, boşluk hareketiyle aynı yönde

$$I_C = I_S e^{V_{BE}/V_T} \Longrightarrow I_B = \left(\frac{I_S}{\beta}\right) e^{V_{BE}/V_T}$$

$$I_D = \frac{I_C}{\beta}$$ → common Emitter current gain

$$I_E = I_C + I_B$$

$$I_C = \beta I_B \quad I_C >> I_B$$

β = ?

$i_E = i_C + i_B$

$I_E + i_e$ ↙ $I_C + i_c$ ↓ $I_B + i_b$ ↘

* p bölgesinde az katkılama yapılır böylece boşluk sayısı az olur ve çok recombinasyonun önüne geçer.

$I_C = \beta I_B$, $I_E = I_C + I_B \Rightarrow I_E = \beta I_B + I_B$

$I_E = \frac{\beta+1}{\beta} I_C \rightarrow \frac{I_C}{\beta}$

$$I_E = \frac{\beta+1}{\beta} I_C$$

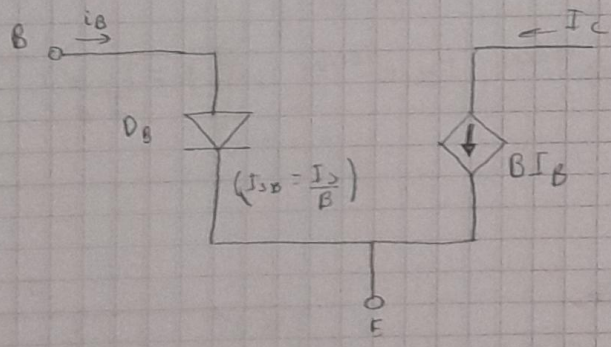
$I_C = I_S e^{V_{BE}/V_T}$

$I_S = \frac{A_E q D_n n_i^2}{N_A \cdot W}$

A_E : Base Emitter Junction Cross sectional Area

W = Effective Base width

Sayfa 313 BJT Large Signal Eqnivalent models in Forword Active



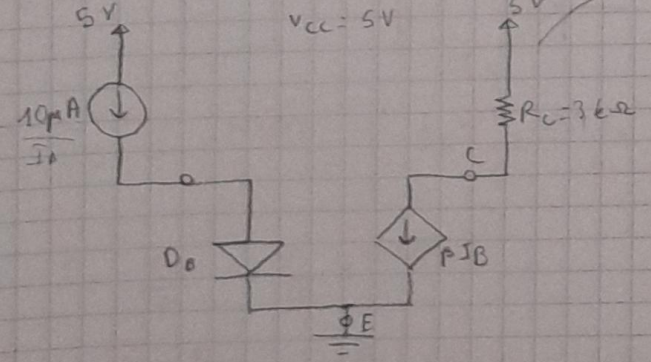
Ex 6.1

npn transistor $V_T = 25mV$

$I_S = 10^{-15} A$, $\beta = 100$

$V_{CC} = 5V$

Active modda → $V_{BE} = ?$ $V_{CE} = ?$ $R_B = ?$



$I_B = \frac{I_S}{\beta} e^{V_{BE}/V_T}$

$\Rightarrow V_{BE} = V_T \ln \frac{I_E}{I_S/\beta}$

$= 25 \ln \left(\frac{10 \cdot 10^{-6}}{10^{-17}} \right)$

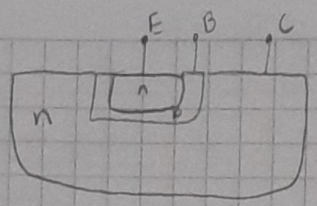
$= 690 mV = 0,69 V$

$I_C = \beta I_B = 100 \times 10 \cdot 10^{-6}$

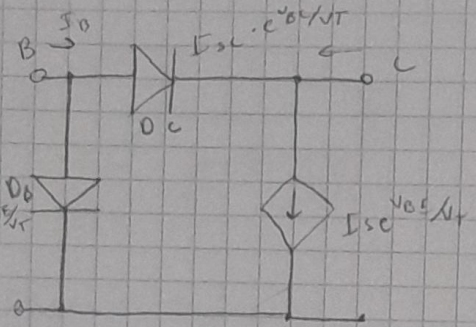
$= 1 mA$

$V_C - V_E = V_{CE} = 5 - 3 \times 1 = 2V$ $V_{CC} - V_C = R_C I_C$

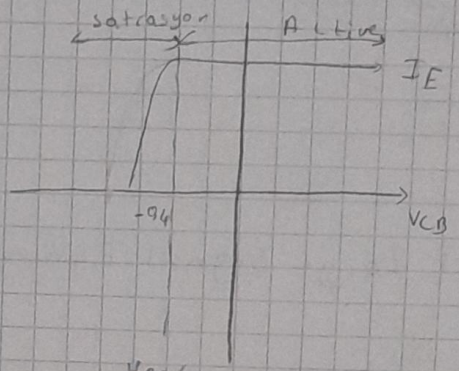
$R_B = \frac{5 - 0,69}{10\mu A} = 431 k\Omega$ aynı koşulları sağlayan direnç kaç olmalıdır akım kaynağı yerine?



Operation in the satraction mode



$I_C = I_S e^{V_{BE}/V_T} - I_{sc} e^{BC/V_T}$



$V_{CE\,sat} = 0,3V$
at the edge of satration

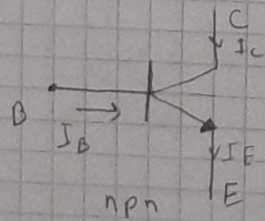
$$I_B = \left(\frac{I_S}{\beta}\right) e^{V_{BE}/V_T} + I_{sc} e^{V_{BC}/V_T} \qquad \beta = \frac{I_C}{I_B}$$

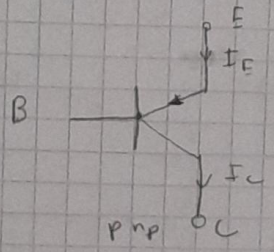
Satrasyon

Forced(sat) $\frac{I_C}{I_B} \leq \beta$

$V_{CE\,sat} = V_{BE} - V_{BC}$

$V_{CE\,sat} \approx 0,1V$ to $0,3V$





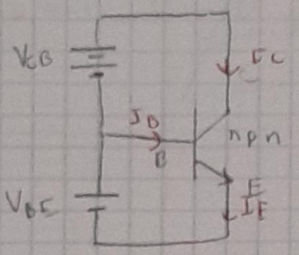
$I_E = I_B + I_C$

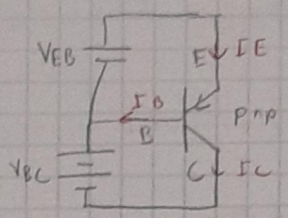
Active $I_C = \beta I_B$

ders 7

BJT Circuits

Sayfa 321





$I_E = I_B + I_C$

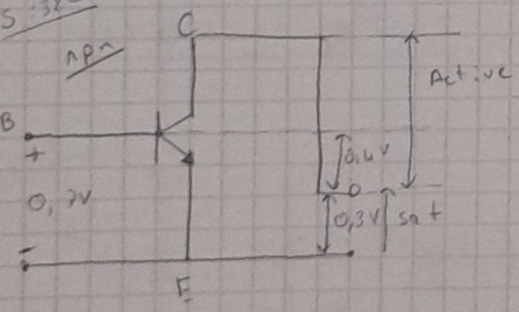
$V_{CE\,sat} = 0,2V$

$I_E = I_B + I_C$

$V_{EC\,sat} = 0,2V$

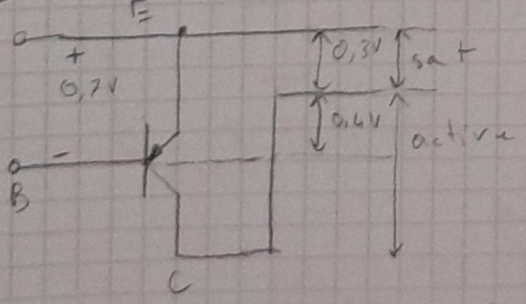
S. 322

npn



Base gerilimi referans alınsa 0,4 V aşağıda düşmesini Aktif mod için de sağlıyor. Daha fazla düşerse satürasyona olur.

pnp

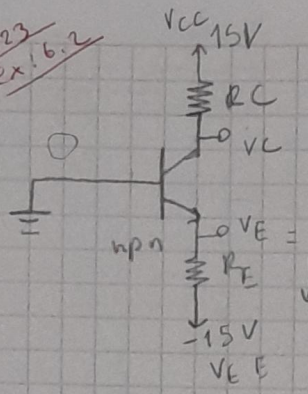


$$\star\quad I_C = I_S e^{V_{BE}/V_T} \qquad I_B = \frac{I_C}{\beta} = \frac{I_S}{\beta} e^{V_{BE}/V_T}$$

$$I_E = \frac{I_C}{\alpha} = \frac{I_S}{\alpha} e^{V_{BE}/V_T}$$

$\alpha = \frac{\beta}{\beta+1}$, $V_T = 25mV$ at room temp

s.323
ex: 6.2



$\beta = 100$

$V_{BE} = 0,7\,V$ @ $1\,mA$ → I_C

Design the circuit ⇒ $I_C = I_{C_1} = 2mA$, $V_C = 5V$

$$R_C = \frac{15 - 5}{2\,mA} = 5\,k\Omega$$

★ $I_{C_1} = I_S\, e^{V_{BE_1}/V_T}$ — 1mA de 0,7V olugorsa

$I_{C_2} = I_S\, e^{V_{BE_2}/V_T}$ — 2mA de kac volt olur.

$$\frac{I_{C_2}}{I_{C_1}} = e^{\frac{V_{BE_2} - V_{BE_1}}{V_T}}$$

$$\Rightarrow V_{BE_2} - V_{BE_1} = V_T \ln\left(\frac{I_{C_2}}{I_{C_1}}\right)$$

$$V_{BE_2} = \underbrace{V_{BE_1}}_{0,7V} + V_T \ln\left(\frac{I_{C_2}}{I_{C_1}}\right) \quad \frac{I_{C_2} \to 2mA}{I_{C_1} \to 1mA}$$

$$\boxed{V_{BE} = 0,717\,V}$$

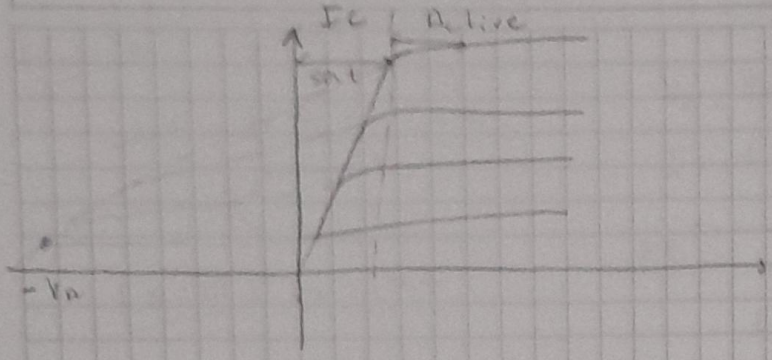
$$I_E = \frac{I_C}{\alpha} = \frac{2}{0,99} = 2,02\,mA$$

$$R_E = \frac{V_E - (-15)}{I_E} = \frac{-0,717 + 15}{2,02} = 7,07\,k\Omega$$

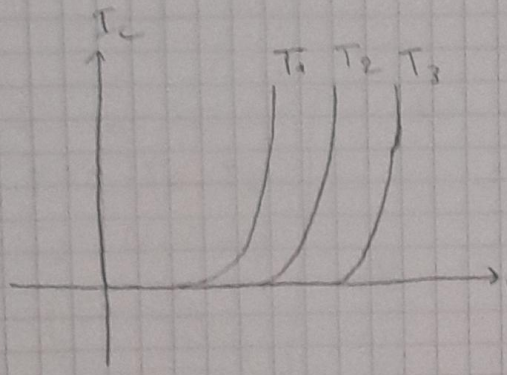
★ $V_{CE} = 5V - (-0,717\,V)$

$= 5,717\,V$ — $V_{CE} \geq 0,3V$ ozaman aktif mod'a

aktif çıkmazsa satfa göre çözmeklazım ve V_{CE} sat'i belirlemek lazım $V_{CE} = 0,2V$

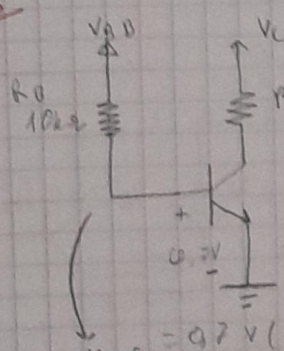


$$I_C = I_{SE}\, e^{V_{BE}/V_T}\left(1+\frac{V_{CE}}{V_A}\right)$$



$T_1 > T_2 > T_3$

Ödev 6.3



$V_{BB} = ?$

a) active mod $V_{CE} = 5V$

b) at the edge of saturation

c) deep saturation $\beta_{forced} = 10$

a) $I_C = \frac{10-5}{1k} = 5mA$

$\rightarrow I_B = \frac{I_C}{\beta} = \frac{5}{50} = 0,1mA$

$V_{BB} = I_B R_B + V_{BE}$

$= (0,1)(10) + 0,7 = \boxed{1,7V}$

★ Sınırda $V_{CE} = 0,3V$
★ Satürasyonda $V_{CE} = 0,2V$

b) at the edge of sat
↳ $V_{CE} = 0,3V$

$I_C = \frac{10-0,3}{1} = 9,7mA$

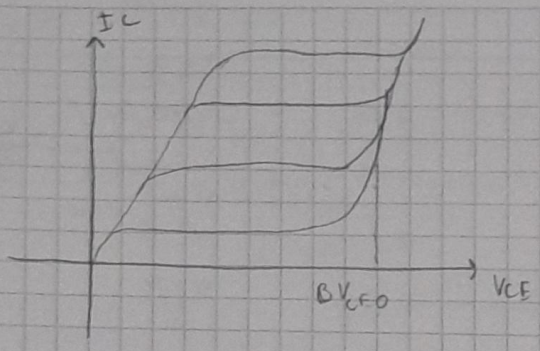
$I_B = \frac{9,7}{50} = 0,194mA$

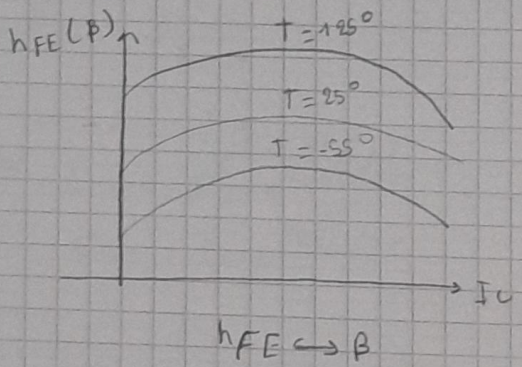
$V_{BB} = (0,194)(10) + 0,7 = 2,64\ V$

c) deep saturation → $\beta_{forced} = 10$
$V_{CE} = V_{CE\ sat} \approx 0,2V$

$I_C = \frac{10-0,2}{1} = 9,8\ mA \Rightarrow I_B = \frac{I_C}{\beta_{forced}} = \frac{9,8}{10} = 0,98mA$

$V_{BB} = (0,98)\times 10 + 0,9 = 10,5V$

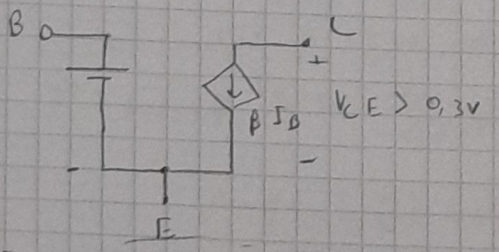




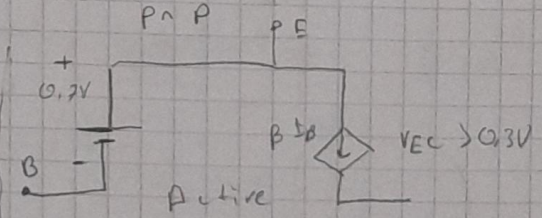
$h_{FE} \hookrightarrow \beta$

3.3 BJT Circuits at DC

npn

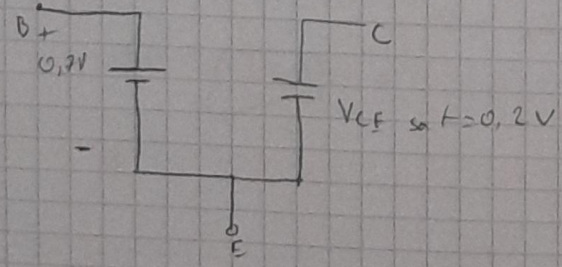


pnp

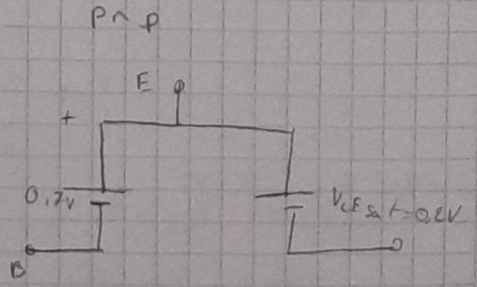


Saturation

npn

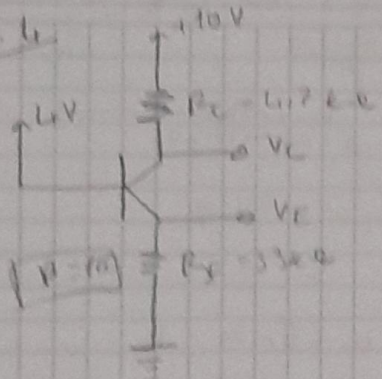


pnp



$I_C \Rightarrow I_B = \frac{I_C}{\beta} \Rightarrow I_E = I_B + I_C$

Ex 6.4



$V_B, V_E, V_C = ?$

$I_B, I_C, I_E = ?$

$V_E = 4 - V_{BE} \approx 4 - 0,7 = 3,3V$

$I_E = \frac{V_E - 0}{R_E} = \frac{3,3}{3,3} = 1mA$

$I_C = I_B + I_E$

$\beta I_B + I_B = I_E$

$I_B(\beta + 1) = I_E$

$I_B = \frac{I_E}{\beta + 1}$

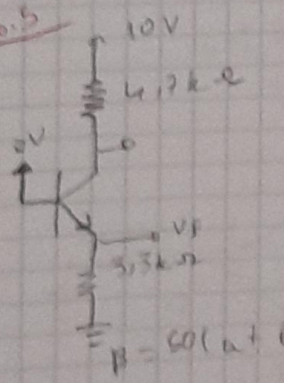
$I_C = \alpha I_E$, $\alpha = \frac{\beta}{\beta+1} = \frac{100}{101} = 0,99$

$I_C = 0,99 \cdot 1 = 0,99mA$

$V_C = 10 - I_C R_C = 10 - 0,99 \cdot 4,7 \approx 5,3V$

$I_B = \frac{I_E}{\beta + 1} = \frac{1}{101} \approx 0,01mA$

Ex 6.5



$V_B, V_C, V_E, I_B, I_C, I_E = ?$

$V_E = 6 - 0,7 = 5,3V$, $I_C = \frac{5,3}{3,3} = 1,6mA$,

$I_E \approx I_C$ $V_C \approx 10 - (1,61)(4,7) = 2,48V$

V_{CE} negatif çıkarsa o zaman yanlış

$\beta = 60$ (at first)

Sonra

$V_E = 6 - 0,7 = 5,3V$

$V_C = 5,5V$

$V_{CE} = 0,2V$ olurdu

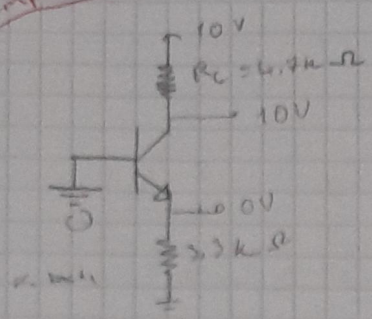
$I_C = \frac{10 - 5,5V}{4,7} = 0,96mA$

$I_E = \frac{5,3}{3,3} = 1,6mA$

$I_B = I_E - I_C = 0,64mA$

★ npn → V_{CE}
pnp → V_{EC}

example 6.6

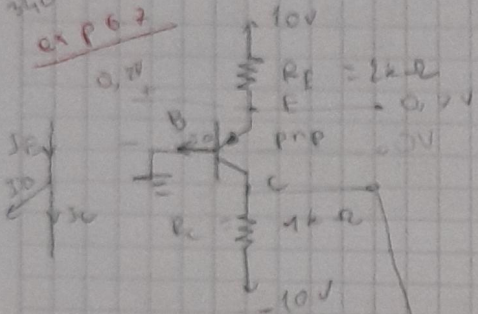


$I_{B,C,E}$, V_B, V_C, V_E

0

3,40

ex p 6.7



$\frac{10 - 0,7}{2} = 4,65\,mA = I_E$ ✓

$\beta = 100$ kabul edelim

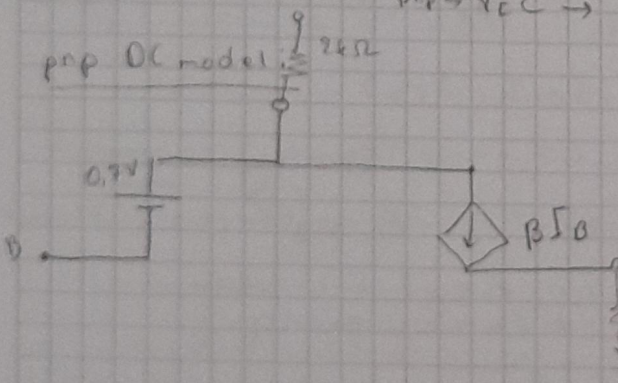
$I_B = \frac{4,65}{10} \approx 0,05\,mA$ ✓

$I_C = 4,6\,mA$ ✓

$-10 + (4,6)(1k) = -5,4$

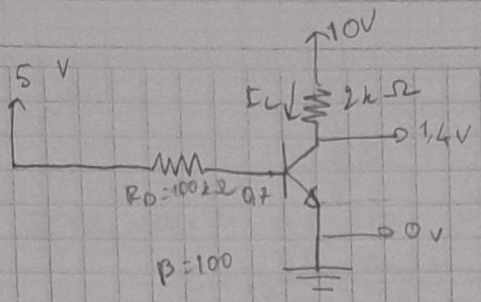
pnp → V_{EC} → $(0,7) - (-5,4) = 6,1\,V$

pnp DC model:



$V_C > 0,3\,V$

Ex 6.8



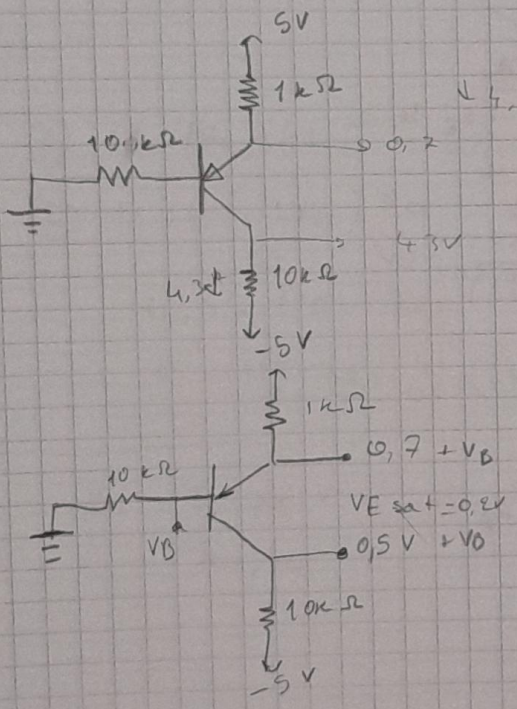
$I_B = \frac{5 - 0.7V}{R_B} = 0,043mA$

$I_C = \beta I_B = (100)(0,043)$
$= 4,3mA$

$V_C = 10 - (I_C)(2)$
$= 10 - (4,3)(2) = 1,4V$

$I_E = (\beta+1) I_B = 101 \times 0,043 \approx 4,3mA$

Ex 6.9



0,7 - 4,3 aktif mod değil

$I_B = \frac{V_B}{10} = 0,1 V_B$

$I_C = \frac{V_B + 0,5 - (-5V)}{10}$
$= \frac{V_B + 5,5}{10} = 0,1 V_B + 0,55$

$I_E = \frac{5 - (V_B + 0,7)}{1k}$
$= 4,3 - V_B$

$4,3 - V_B = (0,1 V_B) + (0,1 V_B) + 0,55)$

$V_B = \frac{3,75}{1,2} \cong 3,13V$

$V_E = 3,73V$

$V_C = 3,63V$

$I_E = 1,17mA$

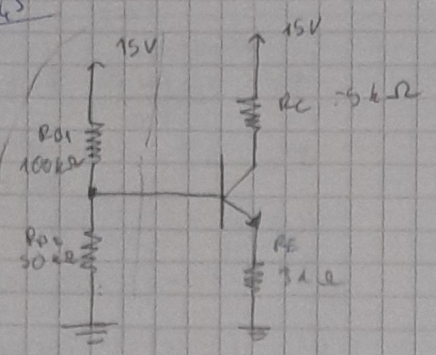
$I_C = 0,86mA$

$I_B = 0,31mA$

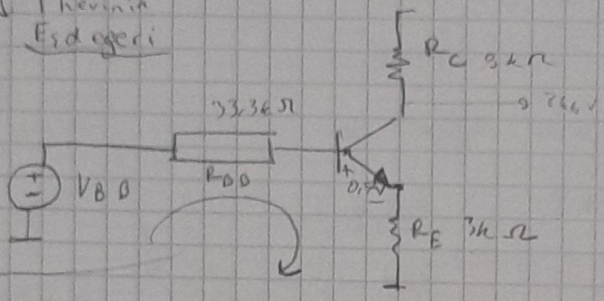
$\beta = \frac{I_C}{I_B} \cong 47$

Ex 6.10

$\beta = 135$



↓ Thevinin Eşdeğeri



$$V_{BB} = 15 \frac{R_{B2}}{R_{B1}+R_{B2}} = 15 \cdot \frac{50}{100+50} = 5V$$

$$R_{BB} = R_{B1} // R_{B2} = 100 // 50 = 33.3\Omega$$

$$V_{BB} = I_B R_{BB} + V_{BE} + I_E R_E$$

$$I_B = \frac{I_E}{\beta+1}$$

$$I_E = \frac{V_{BB} - V_{BE}}{R_E + (R_{BB}/\beta+1)}$$

$$I_E = \frac{5 - 0,7}{3133,3 / 136}$$

$$I_E = 1,29mA$$

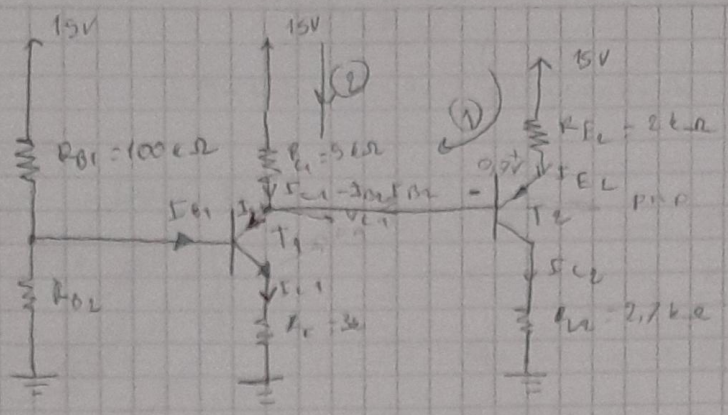
$$I_B = \frac{1,29}{101} = 0,01277mA$$

$$V_B = V_{BE} + I_E R_E = 0,7V + 129,3 = 14,57V$$

$$I_C = \alpha I_E = 0,99 \cdot 129 = 1,28mA$$

$$V_C = 15 - I_C R_C = 15 - (1,28)(3) = 7,16V$$

.11



$I_{B2} = ?$

$V_{C1} = ?$

① $15 - (R_{E2})(I_{E2}) - 0,7 = V_{C1}$

② $15 - (I_{C1} - I_{B2}) R_{C1} = V_{C1}$

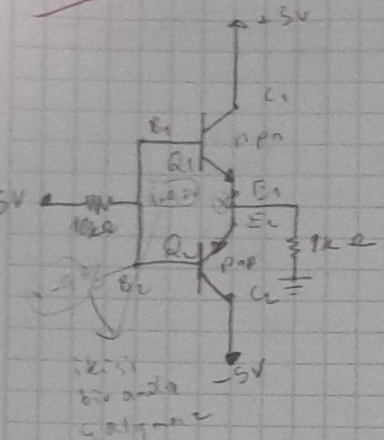
$14,3 - (101) I_{B2} (2k) = V_{C1}$ ①

$15 - (125 - I_{B2}) 5k = V_{C1}$ ②

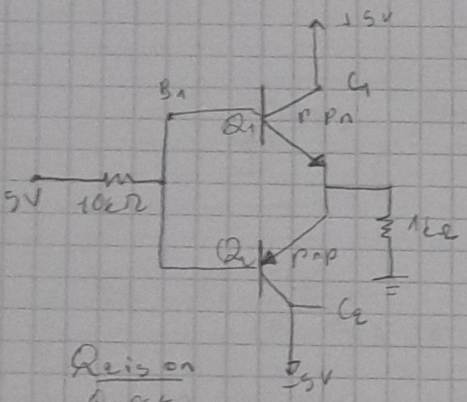
$V_{C1} = 8,738\ V$ $\quad I_{B2} = 27,5\ \mu A$

$\approx 7,74$

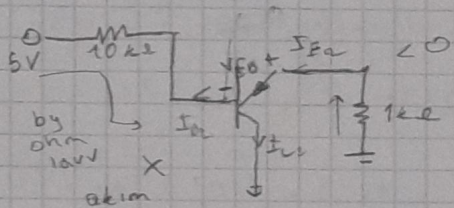
Ex 6.12

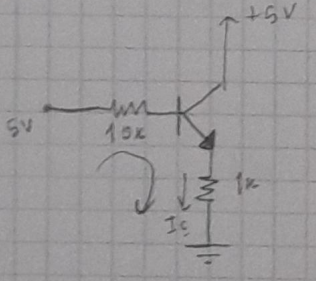


İki si bir arada çalışmaz



Reis on bu ark





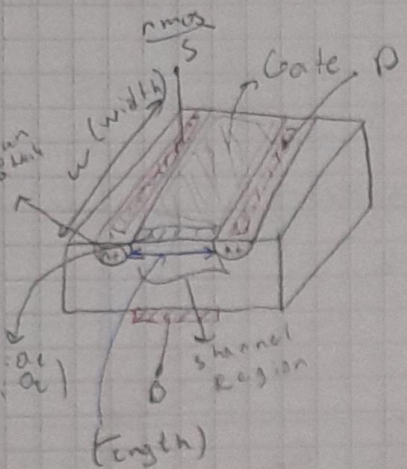
$5V = (10k)\, I_{B1} + 0,7 + (101)\, I_{B1}\,(1k)$

$4,3 = 111k\, I_{B1} \Rightarrow I_{B1} \approx 0,039\ mA$

$\Rightarrow I_{E1} = 101\, I_{B1} \approx 3,9\ mA$

MOSFET

(Metal oxide semiconductor field effect Transistor)



D: Drain → Savak

S: Source → kaynak

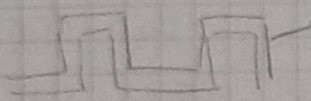
G: Gate → Geçit

B: Body (Bulk) → Gövde

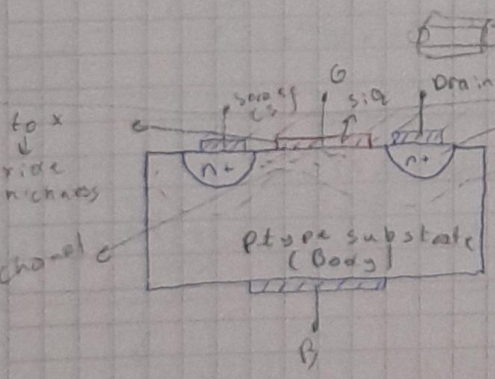
Source ve drain yer değiştirebilir çünkü simetriktir.

Genel bilgi

Modüler



→ Gate all around



Unipolar
tek kutuplu dur

Kullanabilmek için

→ kanal oluşmalı
→ $V_{DS} > 0$

Mosfet türleri: (üretimde kanal şekli)

Enhancement → kanal oluşmalı

Depletion → kanal açık olmalı

Treshold Voltage
(Eşik seviyesi)

↳ V_{GS}'ye ulaşılmalı

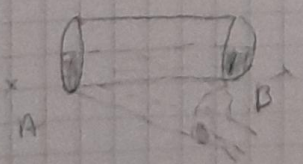
kanal oluşmaya başlar

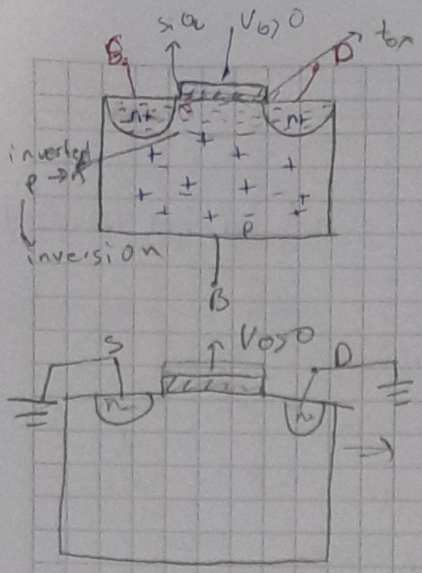
$V_T = \frac{kT}{q}$

farklı

V_t

V_{TH}





$\Rightarrow$ $V_{GS} - V_{TH} = V_{OV}$ → overdrive

$|R| = C_{ox}(WL) \cdot V_{OV}$

C_{ox} = Oxide capacitance of the device

$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}}$ ← elektriksel olanıştır → $\varepsilon_{ox} = (3,9)(\varepsilon_0)$ (ε_r)

$= (3,9)(8,854 \times 10^{-14})$

$= 3,45 \times 10^{-11}$ F/m

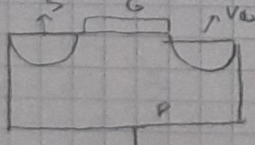
$C_{ox} = \frac{3,45 \times 10^{-11}}{4 \times 10^{-9}} = 7,0 \times 10^{-3}$ F/m^2

*Not

$L = 0,18\,\mu M$ $\quad W = 0,72\,\mu m$

$C = C_{ox} W \cdot L = (8,6)(0,18)(0,72) = 1,1$ fF (10^{-15})

Applying some vds



$|E| = \frac{V_{DS}}{L}$

Elektron drift velocity $= \mu_n \frac{V_{DS}}{L}$ (mobility)

$i_D = \left[\mu_n C_{ox} \left(\frac{W}{L}\right) V_{OV} \right] V_{DS}$

$\mu_n C_{ox}$ → process transconductance parameter

$\left(\frac{W}{L}\right)$ → aspect ratio

V_{OV} → Overdrive voltage $\quad V_{OV} = V_{GS} - V_{TH}$

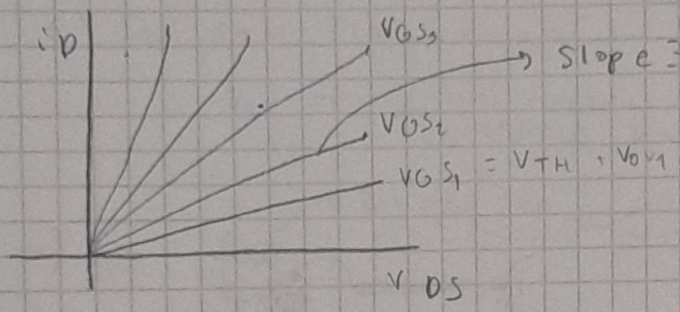
Deep triode Reg. on

$V_{DS} << 2 V_O$

$$g_{DS} = \mu_n C_{ox} \left(\frac{W}{L}\right) V_{OV}$$

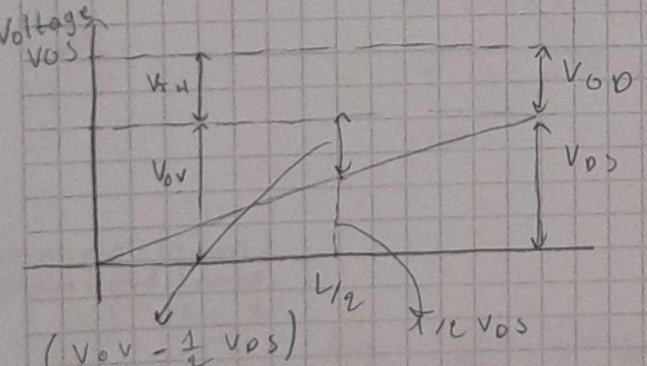
↙ conductance $= \mu_n C_{ox} \left(\frac{W}{L}\right) (V_{GS} - V_{TH})$

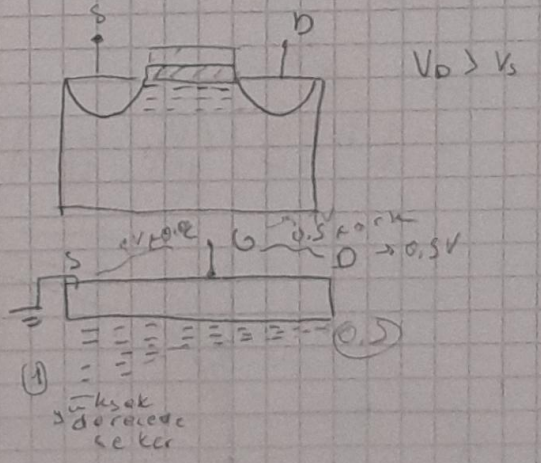
$$r_{DS} = \frac{1}{g_{DS}} = \frac{1}{\mu_n C_{ox} \left(\frac{W}{L}\right) V_{OV}}$$



$$k_n' = \mu_n C_{ox}$$

$$k_n = \mu_n C_{ox} \frac{W}{L} = \beta$$

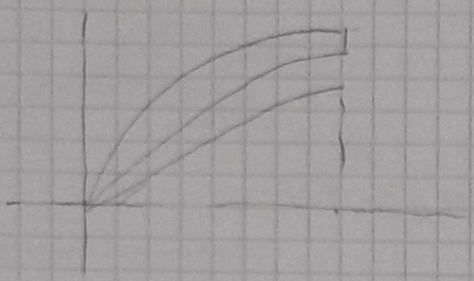




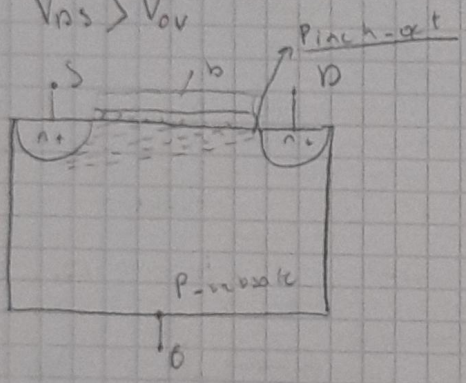
$$i_D = k_n' \left(\frac{W}{L}\right) \left(V_{ov} \cdot V_{DS} - \frac{1}{2} V_{DS}^2\right)$$

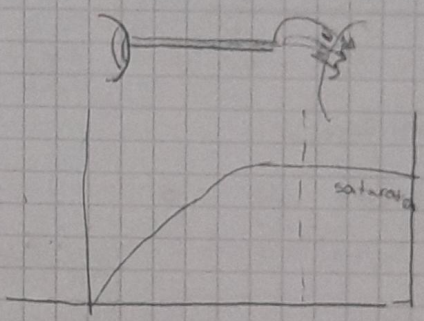
$$\boxed{i_D = k_n' \left(\frac{W}{L}\right) \left[(V_{GS} - V_{TH}) V_{DS} - \frac{1}{2} V_{DS}^2 \right]} \quad \text{(Triode)}$$

Dioda akımı

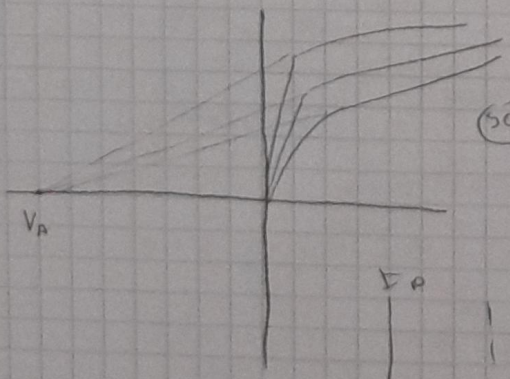


$V_{DS} > V_{ov}$





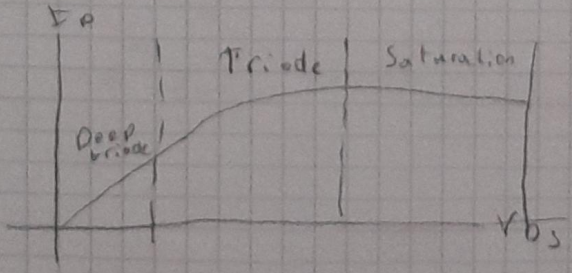
$$i_D = \frac{1}{2} k_n' \left(\frac{W}{L}\right) (V_{GS} - V_{TH})^2$$



Channel length modulation
(kanal boyu modülasyonu)

SOV $\quad V_A = \frac{1}{\lambda}$

$$i_D = \frac{1}{2} k_n' \left(\frac{W}{L}\right) (V_{GS} - V_{TH})^2 (1 + \lambda V_{DS})$$



$$\boxed{i_D = \frac{1}{2} k_n' \left(\frac{W}{L}\right) (V_{GS} - V_{TH})^2}$$

Doyma (saturation)

ex 5.1

$L_{min} = 0,4\,\mu m$ $\quad t_{ox} = 8\,nm$

$\mu_n = 450\,cm^2/V.s$ $\quad V_{TH} = 0,7\,V$

a) Find C_{ox} and k_n'

b) $\frac{W}{L} = \frac{8\,\mu m}{0,8\,\mu m} \rightarrow V_{OV}, V_{GS}, V_{DS\,min}$ → needs to operate the transistor in the sat. region → $I_D = 100\,\mu A$

c) (b) deki transistörün

V_{OV} ve V_{GS} gerilimlerini hesaplayınız

↳ 100Ω Resistor

a) $C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}}$ → $\varepsilon_{ox} = \varepsilon_r \cdot \varepsilon_0$ → 3.9 for SiO2 → $3.45 \cdot 10^{-11}\,F/m$

$= \frac{3.45 \cdot 10^{-11}}{8 \times 10^{-9}} = 4,32 \times 10^{-3}\,F/m^2 = 4,32\,fF/\mu m^2$

$k_n' = \mu_n C_{ox} = (450\,cm^2/V.s)(4.32\,fF/\mu m^2)$

$= 194 \times 10^{-6}\,F/V.s$ → $C = \frac{Q}{V}$ ⇒ $\frac{F}{V.s}$

$= 194\,\mu A/V^2$

$\Rightarrow \frac{Q}{V^2.s} = \frac{I.s}{V^2 s} = \frac{I}{V^2}$

$I = \frac{Q}{s} \rightarrow \frac{dq}{dt} = i$

b) Sat

$I_D = \frac{1}{2} k_n' (V_{OV})^2$

$100 = \frac{1}{2} \times 194 \times \frac{8}{0,8} (V_{OV})^2$

$V_{OV} = 0,32\,V$

$V_{GS} = V_{TH} + V_{OV} = 1,02\,V$

(0,7 + 0,32)

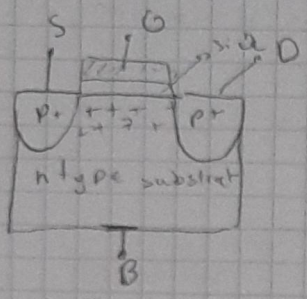
* Sat $V_{DS} \geq V_O$, $V_{DS} \geq V_{GS} - V_{TH}$ saturasyon için

c) Transistor in deep triode region

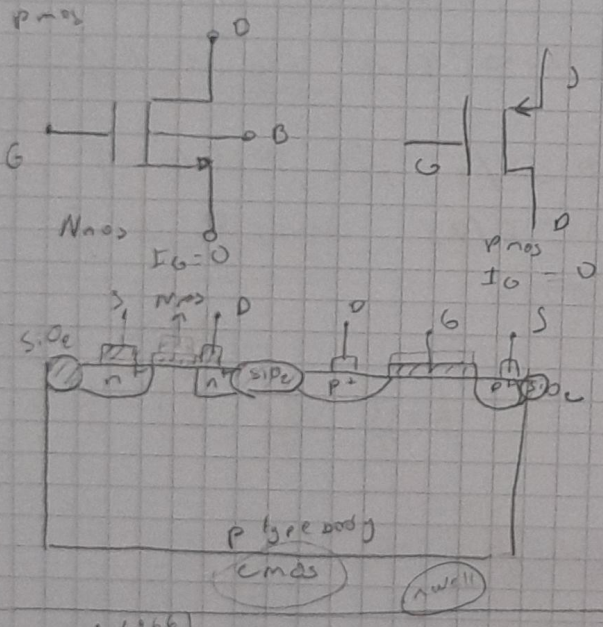
$r_{DS} = \frac{1}{k_n' \frac{W}{L} V_{OV}}$ → $V_{OV} = 0,52V$, $V_{GS} = 1,82V$

r_{DS} ↓ 1000Ω

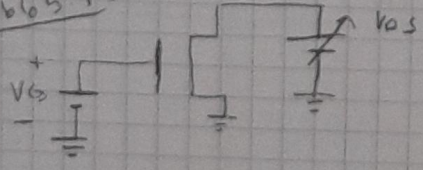
→ sat → ros



pmos + nmos → cmos → complementary mos



Tablos 1 (266)



$V_{GS} < V_{TH}$ → no channel, transistor in cut off ⊗

$V_{GS} \cdot V_{TH} + V_{OV}$ — a channel is induced → triode or saturation

Triode Region
$V_{GD} > V_{TH}$
$V_{DS} < V_{OV}$
$I_D = k_n' \left(\frac{W}{L}\right)\left(V_{OV} - \frac{1}{2}V_{DS}\right)V_{DS}$

Saturation Region
$V_{GD} \leq V_{TH}$
$V_{DS} \geq V_{OV}$
$I_D = \frac{1}{2} k_n' \left(\frac{W}{L}\right)(V_{GS} - V_{TH})^2$

$V_{DS} = V_{GS} - V_{TH}$

$V_D - V_S = V_G - V_S - V_{TH}$

$V_{TH} = G - V_D$